

Theorem Statement

Theorem

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation with associated matrix $A \in \mathbb{R}^{n \times n}$.

1. A is invertible
2. A has a pivot in every row
3. A has a pivot in every column
4. The system $A\mathbf{x} = \mathbf{b}$ has a unique solution for all $\mathbf{b} \in \mathbb{R}^n$
5. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution
6. The columns of A are linearly independent
7. The columns of A span all of $\mathbb{R}^n$.
8. T is invertible
9. T is injective
10. T is surjective

We have the tools to prove all these, but really, it's just a summary of all the linear transformation and matrix algebra we've done so far!

One-sided inverse Implies the Other!

Theorem

Let $A \in \mathbb{R}^{n \times n}$. If there is some $B \in \mathbb{R}^{n \times n}$ such that

$$BA = I_n \text{ or } AB = I_n$$

Then, A is invertible and $A^{-1} = B$.

$AB = I_n$ case

Proof.

First, let $B \in \mathbb{R}^{n \times n}$ such that $AB = I_n$.

We will show that T is surjective by showing how we can get any $\mathbf{b} \in \mathbb{R}^n$.

Let $\mathbf{b} \in \mathbb{R}^n$. See that

$$\mathbf{b} = I_n \mathbf{b} = AB\mathbf{b} = A(B\mathbf{b}) = T(B\mathbf{b})$$

Since we have an input, $\mathbf{x} = B\mathbf{b}$ such that $T(\mathbf{x}) = \mathbf{b}$,

We know that T is surjective, so by above, A is invertible.

Also, see that



$BA = I_n$ case

Proof.

Let $B \in \mathbb{R}^{n \times n}$ such that $BA = I_n$.

We will show that the system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

Assume that $\mathbf{x} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{0}$. See that

$$\begin{aligned} A\mathbf{x} &= \mathbf{0} \\ BA\mathbf{x} &= B\mathbf{0} \\ I_n\mathbf{x} &= \mathbf{0} \\ \mathbf{x} &= \mathbf{0} \end{aligned}$$

So, if $A\mathbf{x} = \mathbf{0}$, then $\mathbf{x}$ must be $\mathbf{0}$!, which means A is invertible.

Next, we show $B = A^{-1}$

$$A^{-1} = I_n A^{-1} = BAA^{-1} = BI_n = B$$

